

MHT-CET 2025 23 April, Shift 1
Question Paper — All 150 Questions with Options

Physics

Chemistry

Mathematics

No.	Subject	Question	(A)	(B)	(C)	(D)
1	Physics	If $\vec{P} = b\hat{i} + 6\hat{j} + \hat{k}$ and $\vec{Q} = \hat{i} - a\hat{j} + 4\hat{k}$ are perpendicular to each other, also $3b - a = 5$. The value of a and b is	$a = 2, b = 1$	$a = 1, b = 2$	$a = 2, b = 3$	$a = 1, b = 3$
2	Physics	Which of the following comes under the category of random errors?	Improper calibration of thermometer	Zero error of $1\mu\text{V}$ in voltmeter	Student measures 22° , where as the correct angle is 20°	Error in measurement of length
3	Physics	The acceleration (a) - time (T) graph for the body starting from rest is given below. The maximum speed of the body is [Figure]	40 m/s	80 m/s	160 m/s	200 m/s
4	Physics	Two bodies of mass 10 kg and 5 kg are moving in concentric circular orbits of radii 'R' and 'r' respectively such that their periods are same. The ratio between their centripetal acceleration is	R/r	r/R	R^2/r^2	r^2/R^2
5	Physics	Radius of curved road is 'R', width of road is 'b'. The outer edge of road is raised by 'h' with respect to inner edge so that a car with velocity 'V' can pass safe over it, then value of 'h' is (g = acceleration due to gravity)	$\frac{V^2 b}{Rg}$	$\frac{V}{Rg}$	$\frac{V^2 R}{g}$	$\frac{V^2}{g}$
6	Physics	The weight of man in a stationary lift is w_1 and when it is moving downwards with uniform acceleration 'a' is w_2 . If the ratio $w_1 : w_2 = 4 : 3$, then the value of 'a' is (g = acceleration due to gravity)	$\frac{g}{3}$	$\frac{g}{4}$	$\frac{3}{4} g$	$\frac{4}{g}$
7	Physics	The figure shows two masses 'm' and 'M' connected by a light string that passes through a small hole 'O' at the centre of the table. Mass 'm' is moved round in a horizontal circle with 'O' as the centre. The frequency with which 'm' should be revolved so that 'M' remains stationary is (g = gravitational acceleration) [Figure]	$\frac{1}{\pi} \sqrt{\frac{ML}{mg}}$	$\frac{1}{2\pi} \sqrt{\frac{Mg}{mL}}$	$\frac{1}{\pi} \sqrt{\frac{Mg}{mL}}$	$\frac{1}{2\pi}$
8	Physics	A uniform solid sphere of mass 'm' and radius 'r' is surrounded by a uniform thin spherical shell of radius '2r' and mass 'm' then the gravitational field	at a distance of 15 r from the centre is $\frac{2}{9} \frac{Gm}{r^2}$	at a distance of $(2.5)r$ from the centre is $\frac{8}{25} \frac{Gm}{r^2}$	at a distance of $(1.5)r$ from the centre is zero.	between two spheres is uniform
9	Physics	A wind with speed 50 m/s blows parallel to the roof of a house. The area of the roof is 300 m^2 . Assume that the pressure inside the house is atmospheric pressure. Density of air is 1.2 kg/m^3 . The magnitude of the force exerted by the wind on the roof will be	$1.5 \times 10^5 \text{ N}$	$3.0 \times 10^5 \text{ N}$	$4.5 \times 10^5 \text{ N}$	$9.0 \times 10^5 \text{ N}$
10	Physics	A disc of paper of radius 'R' has a hole of radius 'r'. It is floating on a liquid of surface tension 'T'. The force of surface tension on the disc is	$2\pi T(R - r)$	$2\pi T(R + r)$	$3\pi T \cdot R$	$4\pi T \cdot R$
11	Physics	The excess pressure inside first soap bubble is three times that of a second soap bubble. The ratio of volumes of the first to second bubble	1 : 3	1 : 9	1 : 27	27 : 1
12	Physics	A thermometer bulb has volume 10^{-6} m^3 and cross-section of the stem is 0.002 cm^2 . The bulb is filled with mercury at 0°C . If the thermometer reads temperature as 100°C , then the length of mercury column is (coefficient of cubical expansion of mercury = $18 \times 10^{-5}/^\circ\text{C}$)	90 cm	9 cm	9 mm	0.9 mm

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No.	Subject	Question	(A)	(B)	(C)	(D)
13	Physics	Two spherical black bodies have radii ' R_1 ' and ' R_2 '. Their surface temperatures are $T_1 K$ and $T_2 K$ respectively. If they radiate the same power, the ratio $\frac{R_1}{R_2}$ is	$\left(\frac{T_1}{T_2}\right)^4$	$\left(\frac{T_1}{T_2}\right)^2$	$\left(\frac{T_2}{T_1}\right)^4$	$\left(\frac{T_2}{T_1}\right)^2$
14	Physics	Two moles of an ideal monoatomic gas undergo a cyclic process as shown in figure. The temperatures in different states are given as $6 T_1 = 3 T_2 = 2 T_4 = T_3 = 2400 K$. The work done by the gas during the complete cycle is (R = Universal gas constant) [Figure]	$-1600R$	$1600R$	$-1200R$	$800R$
15	Physics	A polyatomic gas is compressed to $\left(\frac{1}{8}\right)^{\text{th}}$ of its volume adiabatically. If its initial pressure is P_0 , its new pressure will be [Given, $\frac{C_p}{C_v} = 4/3$]	$6P_0$	$2P_0$	$8P_0$	$16P_0$
16	Physics	The pressure 'P', volume 'V' and temperature 'T' of a gas in a jar 'A' and the gas in other jar 'B' is at pressure ' $2P$ ', volume ' $\frac{V}{4}$ ' and temperature ' $\frac{T}{4}$ '. Then the ratio of the number of molecules in jar A and jar B will be	$1 : 1$	$1 : 2$	$2 : 1$	$4 : 1$
17	Physics	How much should the pressure be increased in order to reduce the volume of a given mass of gas by 5% at the constant temperature?	5%	10%	5.26%	4%
18	Physics	The amplitude of a damped oscillator becomes $\left(\frac{1}{3}\right)^{\text{rd}}$ of original amplitude in 2 seconds. If its amplitude after 6 second become $\left(\frac{1}{n}\right)$ times the original amplitude, the value of n is (n is non zero integer)	9	3	81	27
19	Physics	A particle is executing linear S.H.M. starting from mean position. The ratio of the kinetic energy to the potential energy of the particle at a point of half the amplitude is	$2 : 1$	$3 : 1$	$4 : 1$	$8 : 1$
20	Physics	A particle performs S.H.M. of amplitude 'A' and wavelength ' λ ', Then the velocity of the wave (V) and the maximum particle velocity (ν) are related as	$\nu = \frac{\lambda v}{4\pi A}$	$V = \frac{\lambda \nu}{4\pi^2 A}$	$v = \frac{2\pi A}{\lambda} V$	$V = \frac{\lambda \nu}{4\pi A}$
21	Physics	A pipe open at both ends produces a fundamental frequency n_1 . When the pipe is kept with $\frac{3}{4}^{\text{th}}$ of its length in water, it produces a note of fundamental frequency n_2 . The ratio of $\frac{n_1}{n_2}$ is	$\frac{4}{3}$	$\frac{3}{4}$	2	$\frac{1}{2}$
22	Physics	Two tuning forks when sounded together produce 4 beats per second. One of the forks is in unison with 23 cm length of sonometer wire and other with 24 cm length of the same wire. The frequencies of the two tuning forks are	96 Hz, 92 Hz	92 Hz, 88 Hz	72 Hz, 68 Hz	48 Hz, 44 Hz
23	Physics	An observer moves towards a stationary source of sound with a velocity of one fifth of the velocity of sound. The percentage increase in the apparent frequency is	5%	10%	20%	25%
24	Physics	Two identical straight wires are stretched so as to produce 6 beats per second when vibrating simultaneously with tensions ' T_1 ' and ' T_2 ' respectively. On changing the tension slightly in one of them, the beat frequency remains unchanged. This will happen when (Given $\rightarrow T_1 > T_2$)	T_1 is increased or T_2 is decreased	T_1 is increased by 144	T_2 is decreased by 144	T_1 is decreased or T_2 is increased
25	Physics	Three equal charges are placed on the three corners of a square as shown below. If the magnitude of force between ' q_1 ' and ' q_2 ' is ' F_{12} ' and that between ' q_1 ' and ' q_3 ' is ' F_{13} ', then the ratio of F_{13} to F_{12} is [Figure]	$\frac{1}{\sqrt{2}}$	$\sqrt{2}$	$\frac{1}{2}$	2
26	Physics	A regular hexagon of side 6 cm has a charge of $2\mu C$ at each of its vertices, what is the potential at the centre of the hexagon? $\left[\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI unit}\right]$	$1.5 \times 10^5 V$	$1.8 \times 10^6 V$	$2.4 \times 10^5 V$	$3.2 \times 10^5 V$
27	Physics	The electric field intensity near a conducting surface having uniform surface charge density ' σ ' is given by (ϵ_0 = permittivity of free space)	$\frac{\sigma}{\epsilon_0}$ and is parallel to the surface	$\frac{2\sigma}{\epsilon_0}$ and is parallel to the surface	$\frac{\sigma}{\epsilon_0}$ and is normal to the surface	$\frac{2\sigma}{\epsilon_0}$ and is normal to the surface

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No.	Subject	Question	(A)	(B)	(C)	(D)
28	Physics	A parallel plate air capacitor has capacity ' C ' and distance of separation between plates is ' d '. If a conducting sheet of thickness $\frac{2d}{3}$ is inserted in between the plates, the capacitance becomes C_1 . The ratio of $\frac{C_1}{C}$ is	2 : 1	4 : 1	3 : 1	5 :
29	Physics	In the following circuit, the current through 6Ω resistor is [Figure]	$\frac{1}{5}$ A	$\frac{2}{5}$ A	$\frac{1}{4}$ A	$\frac{3}{4}$ A
30	Physics	The equivalent resistance of the following circuit when no current flows in the resistance of 5Ω is nearly [Figure]	13Ω	17Ω	19Ω	21Ω
31	Physics	Let the current ' I ' be associated with an electron of charge ' e ' moving in a circular orbit of radius ' r ' with speed ' v ' around the positively charged nucleus. The ratio $\frac{r}{v}$ is	$\frac{eI}{2\pi}$	$\frac{2\pi}{eI}$	$\frac{e}{2\pi I}$	$\frac{2eI}{\pi}$
32	Physics	The magnetic field at the centre of a current carrying circular coil of area ' A ' is ' B '. The magnetic moment (M) of the coil is (μ_0 = permeability of free space)	$\frac{BA^2}{\mu_0\pi}$	$\frac{2B}{\mu_0}\frac{A^{3/2}}{\sqrt{\pi}}$	$\frac{BA^{3/2}}{\mu_0\pi}$	$\frac{\mu_0}{BA}$
33	Physics	A square coil ABCD of side ' L ' is carrying a current in clockwise direction. A straight conductor carrying current I_2 (upward direction) is kept parallel to side AB at a distance $\frac{L}{3}$ in the plane of $ABCD$. The net force on the coil ABCD is (μ_0 = magnetic permeability)	$\frac{\mu_0 I_1 I_2}{3\pi}$	$\frac{3\mu_0 I_1 I_2}{4\pi}$	$\frac{3\mu_0 I_1 I_2}{8\pi}$	$\frac{9\mu_0}{8}$
34	Physics	Figure shows the north pole of a magnet moving away from a thick conducting loop containing a capacitor. The excess positive charge will arrive on [Figure]	plate 'a'	plate 'b'	both plates 'a' and 'b'	nei a ' b '
35	Physics	When a current in the conducting coil is changed from 5 A in one direction to 5 A in opposite direction in 0.5 second, an average induced e.m.f in the coil is 2 V . The self-inductance of the coil is	25 mH	50 mH	75 mH	100
36	Physics	Figure shows the combination of inductances and capacitances. Resonant frequency of the L - C circuit is [Figure]	$\frac{1}{2\pi\sqrt{LC}}$	$\frac{1}{3\pi\sqrt{LC}}$	$\frac{1}{4\pi\sqrt{LC}}$	$\frac{1}{6\pi\sqrt{LC}}$
37	Physics	A current of 5 A is flowing at 220 V in a primary coil of a transformer. If the voltage produced in the secondary coil is 2200 V and 50% of power is lost, then the current in the secondary coil will be	2.5 A	5 A	0.25 A	0.0
38	Physics	In an a.c. circuit with pure capacitance 'C' and a.c. source $E = E_0 \sin \omega t$, the equation of instantaneous current is given by	$I = E_0 \omega c \sin(\omega t)$	$I = E_0 \omega c \sin(\omega t + \frac{\pi}{2})$	$I = \frac{E_0}{\omega c} \sin(\omega t)$	$I = \frac{E_0}{\omega c}$
39	Physics	An a. c. source is connected across a pure inductor. Which one of the following figures shows the correct phase relation between the current and e.m.f.? [Figure]	(D)	(B)	(A)	(A)
40	Physics	A ray of light is incident on first face of an equilateral glass prism at an angle of incidence ' i '. The emergent ray just grazes along the adjacent face. The refractive index of the the prism is $\sqrt{2}$. The value of angle of incidence at first face is ($\sin 45^\circ = \frac{1}{\sqrt{2}}, \sin 90^\circ = 1$)	$\sin^{-1}(\sqrt{2} \sin 15^\circ)$	$\sin^{-1}\left(\frac{\sin 30^\circ}{\sqrt{2}}\right)$	$\cos^{-1}(\sqrt{2} \sin 20^\circ)$	$\tan^{-1}(\sqrt{2} \sin 15^\circ)$
41	Physics	Two polaroids are oriented with their planes perpendicular to incident light and transmission axis making an angle 30° with each other. What fraction of incident unpolarised light is transmitted? ($\cos 30^\circ = \sqrt{3}/2$)	$57 \cdot 5\%$	$17 \cdot 5\%$	$27 \cdot 5\%$	$37 \cdot 5\%$
42	Physics	Let the speed of light and the polarising angle for a given medium be ' V ' and ' i_p ' respectively. The relation between them is (C = speed of light in vacuum)	$V = C \cos(i_p)$	$V \cos(i_p) = C \sin(i_p)$	$C = V \cot(i_p)$	$V \sin(i_p) = C \cot(i_p)$
43	Physics	In Young's double slit experiment the wavelength of light used is 6000, the screen is 40 cm from the slits and the fringe width is 0.012 cm , the distance between two slits is	0.024 cm	2.4 cm	0.24 cm	0.2

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No.	Subject	Question	(A)	(B)	(C)	(D)
44	Physics	The energy that should be added to an electron to reduce its de-Broglie wavelength from λ to $\frac{\lambda}{2}$ is n times the initial energy. The value of ' n ' is	4	3	2	1
45	Physics	For two different photosensitive materials having work function ϕ and 2ϕ respectively, are illuminated with light of sufficient energy to emit electrons. If the graph of stopping potential versus frequency is drawn, for these two different photosensitive materials, the ratio of slope of graph for these two materials is	1 : 1	1 : 2	1 : 4	4 :
46	Physics	In hydrogen atom, the energy of electron in first and third orbit is ' E_1 ' and ' E_3 ' respectively. If $E_3 = xE_1$ then the value of x will be	$\frac{1}{9}$	$\frac{1}{64}$	$\frac{1}{27}$	$\frac{1}{8}$
47	Physics	The activity of radioactive sample is measured as N_0 counts per minute at time $t = 0$, and $\frac{N_0}{e}$ counts per minute at time $t = 3$ minute, The activity reduces to half its value in time (in minute)	$\frac{1}{3} \log_e 2$	$3 \log_e 2$	$3 \log_{10} 2$	$\frac{1}{\log_e 2}$
48	Physics	In the case of common emitter transistor amplifier, the ratio of collector current to emitter current is 0.96 , then the current gain (β) of the amplifier is	6	48	24	12
49	Physics	In a common emitter transistor amplifier circuit, the current gain $\alpha = 0.8$. If the base current changes by $3\mu A$ then the collector current changes by	$4\mu A$	$6\mu A$	$12\mu A$	$24\mu A$
50	Physics	Which of the following statements is <u>TRUE</u> for n-type semiconductor?	Holes are majority carriers and trivalent atoms are dopants.	Electrons are majority carriers and trivalent atoms are dopants.	Holes are minority carriers and pentavalent atoms are dopants.	Electrons are minority carriers and pentavalent atoms are dopants.
51	Chemistry	What is the number of moles present in 1 m^3 of any gas at STP?	32.4	54.6	44.6	28.4
52	Chemistry	What is the numerical value of angular momentum for an electron in first orbit of hydrogen atom ? ($h = 6.626 \times 10^{-34} \text{ Js}$)	1.05×10^{-34}	$2 \cdot 10 \times 10^{-34}$	3.16×10^{-34}	4.2×10^{-34}
53	Chemistry	Which from following pair of compounds exhibits isomorphism?	Calcite and aragonite	α -quartz and cristobalite	Sodium nitrate and calcium carbonate	Diamine and fullerenes
54	Chemistry	Which of the following compounds has high lattice enthalpy?	LiCl	NaCl	BeF_2	CaF ₂
55	Chemistry	A container contains equal masses of H_2 , He, CO_2 and Ne at a certain temperature. Which of the following gas exerts maximum partial pressure?	H_2	He	CO_2	Ne
56	Chemistry	Calculate the standard enthalpy change for reaction, $\text{C}_2\text{H}_5\text{OH}_{(\ell)} + 3\text{O}_{2(\text{g})} \rightarrow 2\text{CO}_{2(\text{g})} + 3\text{H}_2\text{O}_{(\ell)}$ If, $\Delta_f H^\circ (\text{C}_2\text{H}_5\text{OH}) = -280 \text{ kJ mol}^{-1}$ $\Delta_f H^\circ (\text{CO}_2) = -390 \text{ kJ mol}^{-1}$ $\Delta_f H^\circ (\text{H}_2\text{O}) = -285 \text{ kJ mol}^{-1}$	-678.00 kJ	$-2033 \cdot 00 \text{ kJ}$	$-1355 \cdot 00 \text{ kJ}$	$-1000 \cdot 00 \text{ kJ}$
57	Chemistry	Calculate work done in isothermal reversible expansion of 1 mole ideal gas from initial pressure 10 bar to final pressure 1 bar at 300 K ($R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$)	-2.87 kJ	$-8 \cdot 60 \text{ kJ}$	-5.74 kJ	$-1 \cdot 37 \text{ kJ}$
58	Chemistry	Which from following reactions is exothermic?	$\text{KOH}_{(\text{aq})} + \text{HNO}_{3(\text{aq})} \rightarrow \text{KNO}_{3(\text{aq})} + \text{H}_2\text{O}$	$\text{H}_2\text{O}_{(\text{s})} \rightarrow \text{H}_2\text{O}_{(\text{l})}$	$\text{NaCl}_{(\text{s})} + \text{aq} \rightarrow \text{Na}^+_{(\text{aq})} + \text{Cl}^-_{(\text{aq})}$	$\text{N}_2 + 2\text{O}_2 \rightarrow 2\text{NO}_2$

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No.	Subject	Question	(A)	(B)	(C)	(D)
59	Chemistry	Solubility of binary sparingly soluble salt is $1.12 \times 10^{-4} \text{ g/dm}^3$. Calculate its solubility product (molar mass of salt = 112 g mol^{-1})	1×10^{-3}	1×10^{-6}	1×10^{-9}	1×10^{-12}
60	Chemistry	A weak base is 5% dissociated in its 0.01 M solution. Calculate the dissociation constant.	3.5×10^{-6}	2.0×10^{-5}	2.3×10^{-4}	2.5×10^{-3}
61	Chemistry	Which among the following salts forms basic solution in water?	NH_4NO_3	Na_2CO_3	NH_4Cl	Na_2SO_4
62	Chemistry	Identify correct statement for reaction stated below. $\text{Ag}_2\text{O} + \text{H}_2\text{O} + 2\text{e}^- \rightarrow 2\text{Ag} + 2\text{OH}^-$	Water is oxidised.	Hydrogen is oxidised.	Silver is reduced.	Hydrogen is reduced.
63	Chemistry	Identify the element forming peroxide on reaction with oxygen.	C	Ar	Na	He
64	Chemistry	Which among the following is a pair of functional isomers?	Butan-2-ol and 2-Methylpropanol	Butan-1-ol and 1-Methoxypropane	1-Methoxypropane and Ethoxyethane	2-Methoxypropane and 2-Methoxyethane
65	Chemistry	What is the molar mass of third homologue of alkane series?	40 g mol^{-1}	58 g mol^{-1}	44 g mol^{-1}	46 g mol^{-1}
66	Chemistry	Which of the following compounds forms 1-bromo-1-methylcyclohexane when treated with HBr ?	[Figure]	[Figure]	[Figure]	[Figure]
67	Chemistry	Identify product 'B' in the following sequence of reactions. $1,2 \text{ Dibromoethane} \xrightarrow[\text{alcohol}]{\text{KOH}} \text{A} \xrightarrow{\text{sodamide}} \text{B}$	Bromoethane	Bromoethene	Ethene	Ethanol
68	Chemistry	Calculate the volume occupied by all particles in bcc unit cell if the volume of unit cell is $8.0 \times 10^{-23} \text{ cm}^3$	$3.19 \times 10^{-23} \text{ cm}^3$	$2.72 \times 10^{-23} \text{ cm}^3$	$5.44 \times 10^{-23} \text{ cm}^3$	$1.4 \times 10^{-23} \text{ cm}^3$
69	Chemistry	Calculate the number of atoms in 1 g metal if it forms fcc crystal structure $[\rho(\text{density}) \times a^3 = 1.728 \times 10^{-22} \text{ g}]$	2.315×10^{22}	3.156×10^{22}	4.108×10^{22}	1.4×10^{22}
70	Chemistry	Identify the principle suggesting to avoid protection and deprotection of selective group for organic reactions in green chemistry.	Reduce derivatives	Use of renewable feed stock	Design for energy efficiency	Design for safety and health
71	Chemistry	Calculate the molality of solution of nonvolatile solute if boiling point elevation is 1.75 K [K_b for solvent = 3 K kg mol^{-1}]	0.480 m	0.33 m	0.58 m	0.6 m
72	Chemistry	Calculate van't Hoff factor of aqueous solution of 0.18 m electrolyte that freezes at -0.54°C (K_f for solvent = $1.86 \text{ K kg mol}^{-1}$)	1.126	2.449	1.612	2.449
73	Chemistry	Determine the expected value of ΔT_f for 1 m CaCl_2 solution if 1 m urea solution has ΔT_f value ' x ' K .	$x \text{ K}$	$2x \text{ K}$	$3x \text{ K}$	$\frac{x}{2} \text{ K}$
74	Chemistry	What is value of E_{cell} at 298 K for the reaction, $\text{Zn}_{(\text{s})} + \text{Cu}^{+2}(0.1\text{M}) \rightarrow \text{Zn}^{+2}(0.1\text{M}) + \text{Cu}_{(\text{s})}$ if $E_{\text{cell}}^\circ = 1.1 \text{ V}$?	1.1 V	0.11 V	1.0408 V	0.011 V
75	Chemistry	Standard potential of electrode reaction $\text{Cu}^{+2}_{(\text{aq})} + 2\text{e}^- \rightarrow \text{Cu}_{(\text{s})}$ is $+0.34 \text{ V}$. What is standard potential of reaction $2\text{Cu}_{(\text{s})} \rightarrow 2\text{Cu}^{+2}_{(\text{aq})} + 4\text{e}^-$?	$+0.68 \text{ V}$	-0.68 V	$+0.34 \text{ V}$	-0.34 V
76	Chemistry	Consider galvanic cell, $\text{A}_{(\text{s})} \left \text{A}^{+2}_{(\text{aq})} \right \left \text{B}^{+}_{(\text{aq})} \right \text{B}_{(\text{s})}^{\oplus}$ if emf of cell is positive, identify a correct cell reaction from following.	$\text{A}_{(\text{s})} + \text{B}^{+2}_{(\text{aq})} \rightarrow \text{A}^{+2}_{(\text{aq})} + \text{B}_{(\text{s})}$	$\text{A}_{(\text{s})} + 2 \text{B}^{+}_{(\text{aq})} \rightarrow \text{A}^{+2}_{(\text{aq})} + 2 \text{B}_{(\text{s})}$	$2 \text{A}^{+2}_{(\text{aq})} + \text{B}_{(\text{s})} \rightarrow 2 \text{A}_{(\text{s})} + \text{B}^{+}_{(\text{aq})}$	$2 \text{A}_{(\text{s})} + \text{B}^{+}_{(\text{aq})} \rightarrow 2 \text{A}^{+2}_{(\text{aq})} + \text{B}_{(\text{s})}$
77	Chemistry	Which of the following is an example of first order reaction?	$\text{CH}_3\text{CHO}_{(\text{g})} \rightarrow \text{CH}_4_{(\text{g})} + \text{CO}_{(\text{g})}$	$2\text{NO}_{2(\text{g})} + \text{F}_{2(\text{g})} \rightarrow 2\text{NO}_{2(\text{g})} + \text{F}_{(\text{g})}$	$2\text{NO}_{(\text{g})} + 2\text{H}_{2(\text{g})} \rightarrow \text{N}_{2(\text{g})} + 2\text{H}_2\text{O}_{(\text{g})}$	$2\text{H}_{2(\text{g})} + \text{O}_{2(\text{g})} \rightarrow 2\text{H}_2\text{O}_{(\text{g})}$
78	Chemistry	The rate constant of a first order reaction is $1.15 \times 10^{-3} \text{ s}^{-1}$. How long will 5 g of reactant take to reduce to 3 g ?	314 second	240 second	404 second	444 second

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79	Chemistry	Find the rate of following reaction. $2 \text{N}_2\text{O}_5(\text{g}) \rightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ if concentration of NO_2 increases to $5.2 \times 10^{-3}\text{M}$ in 100 seconds.	$1.3 \times 10^{-3} \text{ mol dm}^{-3}$	$1.3 \times 10^{-5} \text{ mol dm}^{-3}$	$4.0 \times 10^{-5} \text{ mol dm}^{-3}$	$5.2 \times 10^{-3} \text{ mol dm}^{-3}$
80	Chemistry	Which from following is classified as multimolecular colloid?	Starch	Cellulose	S_8 sulphur molecules	Plasma
81	Chemistry	Identify from following compounds that valence shell of 'Xe' consists one lone pair of electrons.	XeF_2	XeF_4	XeOF_2	XeO_2
82	Chemistry	Which of the following inert gases is used for magnetic resonance imaging?	He	Ne	Ar	Xe
83	Chemistry	Which element from following has largest ionic size in +3 state ?	Pr	Sm	La	Yb
84	Chemistry	Find the value of spin only magnetic moment for Zn^{2+} in BM .	1.73	2.84	0.00	3.87
85	Chemistry	What is EAN of metal ion in hexacyanoferrate (II) ion?	36	38	37	35
86	Chemistry	Which among the following ligands has highest field strength?	H_2O	OH^-	$\text{C}_2\text{O}_4^{2-}$	CO
87	Chemistry	The reaction of aryl halide with sodium metal in dry ether to give biphenyl is known as	Swartz reaction	Wurtz reaction	Fittig reaction	Wittig reaction
88	Chemistry	Identify the final product obtained when a mixture of vapours of phenol and hydrogen is passed over nickel catalyst at 433 K .	Cyclohexanol	Cyclohex-2-en-1-ol	Cyclohexane	Benzene
89	Chemistry	What is the medicinal property of Eugenol?	Antiseptic	Antioxidant	Analgesic and antimicrobial	Anesthetic
90	Chemistry	Which of the following has highest solubility in water?	Phenol	p-Cresol	o-Nitrophenol	p-Nitrophenol
91	Chemistry	Identify the product 'Z' in the following series of reactions. Ethanol [Figure]	Ethylchloride	Ethyl magnesium chloride	Ethyl amine	Ethylamine
92	Chemistry	Which among the following compounds has highest boiling point?	Propanal	Ethanal	Pentanal	Hexanal
93	Chemistry	Identify substrate 'S' in the following reaction. $\text{S} \xrightarrow{\text{Dimethyl cadmium}} \text{Propanone} + \text{Cadmium chloride}$	Ethyl chloride	Ethylene dichloride	Ethanoyl chloride	Ethyl dichloride
94	Chemistry	Identify the reagent 'R' necessary to bring the following conversion. [Figure]	H_2O	CrO_3	dil. NaOH	KMnO_4
95	Chemistry	Which from following is a correct increasing order of basic strength of compounds?	$(\text{CH}_3)_2\text{NH} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_3\text{N} < \text{NH}_3$	$\text{NH}_3 < (\text{CH}_3)_2\text{NH} < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_3\text{N}$	$\text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH} < (\text{CH}_3)_3\text{N} < \text{NH}_3$	$\text{NH}_3 < \text{CH}_3\text{NH}_2 < (\text{CH}_3)_2\text{NH} < (\text{CH}_3)_3\text{N}$
96	Chemistry	What is the number of moles of hydrogen atoms required to reduce one mole of nitroethane completely using Sn and HCl ?	2	3	4	6
97	Chemistry	What type of glycosidic linkages are present in cellulose?	$\alpha - 1, 4$	$\beta - 1, 4$	$\alpha - 1, 6$	$\beta - 1, 6$
98	Chemistry	What is the total number of -OH groups present in ribonucleoside?	5	4	3	2
99	Chemistry	What are the monomers used in preparation of PHBV?	$\text{HOOC} - (\text{CH}_2)_4 - \text{COOH}$ and $\text{H}_2\text{N} - (\text{CH}_2)_6 - \text{NH}_2$	[Figure]	[Figure]	$\text{C}_6\text{H}_5\text{COOH}$
100	Chemistry	Which from following polymers is used to obtain oil seals?	Teflon	Dacron	Nylon 6	LDPE
101	Mathematics	If $x = -2 + \sqrt{-3}$, then the value of $2x^4 + 5x^3 + 7x^2 - x + 38$ is equal to	1	-2	3	5

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No.	Subject	Question	(A)	(B)	(C)	(D)
102	Mathematics	A family consisting of a mother, father and their 8 children (4 boys and 4 girls) are to be seated at a round table in a party. How many ways can this be done if the mother and father sit together and the males and females alternate?	567	765	657	576
103	Mathematics	If $\sin A + \sin B = x$ and $\cos A + \cos B = y$, then $\sin(A+B) =$	$\frac{2xy}{x^2+y^2}$	$\frac{xy}{x^2+y^2}$	$\frac{2xy}{y^2-x^2}$	$\frac{x}{y^2-x^2}$
104	Mathematics	The distance of the point $(1, 2)$ from the line $x + y = 0$ measured parallel to the line $3x - y = 2$ is	$\frac{3\sqrt{2}}{8}$ units	$\frac{3\sqrt{10}}{4}$ units	10 units	$5\sqrt{2}$
105	Mathematics	The joint equation of two lines passing through $(-2, 3)$ and parallel to the bisectors of the angle between the co-ordinate axes is	$x^2 - y^2 + 4x + 6y - 4 = 0$	$x^2 + y^2 + 4x + 6y - 5 = 0$	$x^2 - y^2 + 4x + 6y - 5 = 0$	$x^2 + y^2 + 4x + 6y - 4 = 0$
106	Mathematics	Let the circle with centre at origin pass through the vertices of an equilateral triangle ABC . If $A \equiv (2, 4)$, then the length of the median through A is	$2\sqrt{5}$ units	$3\sqrt{5}$ units	$4\sqrt{5}$ units	$6\sqrt{5}$ units
107	Mathematics	If the tangent at the point $(2 \sec \theta, 3 \tan \theta)$ to the hyperbola $\frac{x^2}{4} - \frac{y^2}{9} = 1$ is parallel to $3x - y + 4 = 0$, then the value of θ is	45°	60°	30°	90°
108	Mathematics	If $f(x) = \frac{10^x + 7^x - 14^x - 5^x}{1 - \cos x}$, $x \neq 0$ is continuous at $x = 0$, then the value of $f(0)$ is	$\log 2 \left[\log \left(\frac{5}{7} \right) \right]$	$\log 4 \left[\log \left(\frac{5}{7} \right) \right]$	$\log 2 \left[\log \left(\frac{7}{5} \right) \right]$	$\log 4 \left[\log \left(\frac{7}{5} \right) \right]$
109	Mathematics	$\lim_{x \rightarrow 2} \frac{x+3x^2+5x^3+7x^4-166}{x-2} =$	167	267	287	297
110	Mathematics	The contrapositive of the statement $\sim p \vee (q \wedge \sim r)$ is	$p \rightarrow q \wedge r$	$(q \wedge r) \rightarrow p$	$\sim q \vee \sim r \rightarrow p$	$(r \vee p) \rightarrow q$
111	Mathematics	Which of the following is the negation of the statement "For all $M > 0$, there exist $x \in s$ such that $x \geq M$ "	$\exists M > 0$ such that $x \geq M$ for all $x \in s$	$\exists M > 0, \exists x \in s$ such that $x \geq M$	$\exists M > 0$ such that $x < M$ for all $x \in s$	$\exists M > 0$ such that $x < M$ for all $x \in s$
112	Mathematics	In a triangle ABC , with usual notations, if $a = 5, b = 4, \cos(A-B) = \frac{31}{32}$, then $c =$	6	7	5	2
113	Mathematics	With usual notations, in $\triangle ABC$, the lengths of two sides are 10 cm and 9 cm respectively. If angles A, B, C are in A.P. then perimeter of ABC is	$24 + 2\sqrt{6}$ cm	$24 + \sqrt{6}$ cm	$24 - 2\sqrt{6}$ cm	$24 - \sqrt{6}$ cm
114	Mathematics	In a triangle ABC with usual notations. $\cot \frac{A}{2} + \cot \frac{B}{2} + \cot \frac{C}{2} =$	$\frac{s^2}{\Delta}$, where Δ is the area of the triangle ABC .	$\frac{s}{\Delta}$, where Δ is the area of the triangle ABC .	$\frac{\Delta}{s}$, where Δ is the area of the triangle ABC .	Δ , where Δ is the area of the triangle ABC .
115	Mathematics	In $\triangle ABC$, with usual notations, if $\cos \frac{B}{2} = \sqrt{\frac{c+a}{2a}}$, then $a^2 =$	$b^2 - c^2$	$b + c$	$b^2 + c^2$	$b - c$
116	Mathematics	If A and B are non-singular matrices of order 2 such that $(AB)^{-1} = \frac{1}{6} \begin{bmatrix} -7 & -3 \\ 2 & 3 \end{bmatrix}$ and $A^{-1} = \frac{1}{3} \begin{bmatrix} 4 & 3 \\ -1 & 0 \end{bmatrix}$ then $B^{-1} =$	$\frac{1}{2} \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$	$\frac{1}{2} \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$	$\frac{1}{2} \begin{bmatrix} -1 & 3 \\ 1 & 2 \end{bmatrix}$	$\frac{1}{6} \begin{bmatrix} -7 & -3 \\ 2 & 3 \end{bmatrix}$
117	Mathematics	The value of $\tan^2(\sec^{-1} 4) + \cot^2(\operatorname{cosec}^{-1} 3)$ is	15	25	23	7
118	Mathematics	If $f(x) = 3x + 10$, $g(x) = x^2 - 1$, then $(f \circ g)^{-1}(x) =$	$\left(\frac{x-7}{3} \right)$	$\left(\frac{x-7}{3} \right)^{\frac{1}{2}}$	$\left(\frac{x-7}{3} \right)^{\frac{1}{3}}$	$\left(\frac{x-7}{3} \right)^{\frac{1}{4}}$
119	Mathematics	If $y = \sqrt{x + \sqrt{y + \sqrt{x + \sqrt{y + \dots \infty}}}}$, then $\frac{dy}{dx} =$	$2y - 1$	$\frac{1}{2y-1}$	$\frac{y^2-x}{2y^3-2xy-1}$	$14C$
120	Mathematics	If $x = a \sin 2t(1 + \cos 2t)$, $y = b \cos 2t(1 - \cos 2t)$ then $\frac{dy}{dx}$ is equal to	$\frac{b}{a} \tan t$	$\frac{a}{b} \tan t$	$\frac{b}{a \tan t}$	$\frac{a}{b \tan t}$
121	Mathematics	If $y = \tan^{-1} \left[\frac{4 \sin 2x}{\cos 2x - 6 \sin^2 x} \right]$, then $\frac{dy}{dx}$ at $x = 0$ is	$\frac{1}{8}$	-8	8	$-\frac{1}{8}$

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No.	Subject	Question	(A)	(B)	(C)	(D)
122	Mathematics	The equation of motion of the particle is $s = at^2 + bt + c$. If the displacement after 1 second is 20 m, velocity after 2 seconds is 30 m/seconds and the acceleration is 10 m/seconds ² , then	$a + c = 2b$	$a + c = b$	$a - c = b$	$a + c = b$
123	Mathematics	A particle moves along a curve $y = \frac{2x^3-1}{3}$. The points on the curve at which the y co-ordinate is changing 18 times the x co-ordinate are	$(-3, -\frac{55}{3}), (3, -\frac{55}{3})$	$(-3, \frac{53}{3}), (3, \frac{53}{3})$	$(-3, -\frac{53}{3}), (3, \frac{53}{3})$	$(-3, -\frac{53}{3}), (3, \frac{53}{3})$
124	Mathematics	The point on the curve $4y^2 - 4y + 2x - 1 = 0$ at which the tangent becomes parallel to Y-axis is	$(1, \frac{1}{2})$	$(\frac{1}{2}, 1)$	$(-1, -\frac{1}{2})$	$(\frac{1}{2}, -1)$
125	Mathematics	The maximum value of $x^{2/3} + (x-2)^{2/3}$ is	0	2	$2\frac{2}{3}$	1
126	Mathematics	$\int \left(\frac{x-3}{x^2+9}\right)^2 dx =$	$\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) - \frac{3}{x^2+9} + c$, where c is the constant of integration.	$\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) - \frac{1}{x^2+9} + c$, where c is the constant of integration.	$\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + \frac{3}{x^2+9} + c$, where c is the constant of integration.	$\frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + \frac{1}{x^2+9} + c$, where c is the constant of integration.
127	Mathematics	$\int \frac{dx}{e^x-1} =$	$\log(e^x - 1) + x + c$, where c is the constant of integration.	$\log(e^x - 1) - x + c$, where c is the constant of integration.	$x - \log(e^x - 1) + c$, where c is the constant of integration.	$\log(e^x - 1) + x + c$, where c is the constant of integration.
128	Mathematics	$\int \frac{dx}{\sqrt{x}+x} =$	$\log \sqrt{x} + c$, where c is the constant of integration.	$\log(\sqrt{x} + x) + c$, where c is the constant of integration.	$\log(1 + \sqrt{x}) + c$, where c is the constant of integration.	$2 \log \sqrt{x} + c$, where c is the constant of integration.
129	Mathematics	$\int_3^5 \frac{\sqrt{x} dx}{\sqrt{8-x} + \sqrt{x}} =$	0	1	2	3
130	Mathematics	The value of $\int_0^1 \tan^{-1}(1-x+x^2) dx$ is	$\frac{\pi}{2} - \log 2$	$\frac{\pi}{2} + \log 2$	$\log 2$	0
131	Mathematics	The area bounded by the curve $x^2 = 8y$ and the straight line $x - 8y + 2 = 0$ is	$\frac{9}{8}$ sq. units	$\frac{15}{16}$ sq. units	$\frac{9}{16}$ sq. units	$\frac{15}{8}$ sq. units
132	Mathematics	The differential equation of all circles touching the Y-axis at the origin and centre on the X-axis is	$x^2 + y^2 + 2xy \frac{dy}{dx} = 0$	$x^2 - y^2 + 2xy \frac{dy}{dx} = 0$	$2x^2 + y^2 + xy \frac{dy}{dx} = 0$	$x^2 - y^2 + xy \frac{dy}{dx} = 0$
133	Mathematics	The order and degree of the differential equation $\sqrt{\frac{dy}{dx}} - 4 \frac{dy}{dx} - 7x = 0$ is respectively	1,2	2,1	2,2	3,1
134	Mathematics	If $x \frac{dy}{dx} = y(\log y - \log x + 1)$, then the solution of the equation is	$\log \frac{x}{y} = cy$, where c is the constant of integration	$\log \frac{y}{x} = cy$, where c is the constant of integration	$\log \frac{x}{y} = cx$, where c is the constant of integration	$\log \frac{y}{x} = cx$, where c is the constant of integration
135	Mathematics	The rate at which a substance cools in moving air, is proportional to the difference between the temperature of the substance and that of air. The temperature of air is 290 K and the substance cools from 370 K to 330 K in 10 minutes. Then the time to cool the substance upto 295 K is	40 min	96 min	50 min	60 min
136	Mathematics	Let $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and vector $\vec{c}$ be coplanar. If $\vec{c}$ is perpendicular to $\vec{a}$, then $\vec{c}$ is	$-\hat{i} + 2\hat{k}$	$-\hat{i} + \hat{j} + \hat{k}$	$\hat{i} - 2\hat{j}$	$-\hat{j}$
137	Mathematics	If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $ \vec{a} = \sqrt{31}, \vec{b} = \vec{c} = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$ and if the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$ then $\left \frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right ^2 =$	1	2	3	11
138	Mathematics	The area of the rectangle having vertices P, Q, R, S with position vectors $-\hat{i} + \hat{j} + \hat{k}, \hat{i} + \hat{j} + \hat{k}, \hat{i} - \hat{j} + \hat{k}, -\hat{i} - \hat{j} + \hat{k}$ respectively is	1 square unit	2 square units	3 square units	4 square units
139	Mathematics	Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b}$ and $\vec{c} = \hat{j} - \hat{k}$ be three vectors such that $\vec{a} \times \vec{b} = \vec{c}$ and $\vec{a} \cdot \vec{c} = 0$. If the length of projection vector of the vector $\vec{b}$ on the vector $\vec{a} \times \vec{c}$ is l , then the value of $3l^2$ is	1	2	4	6

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No.	Subject	Question	(A)	(B)	(C)	(D)
140	Mathematics	Let $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are vectors such that $\vec{a} \times \vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{c} \times \vec{d} = 3\hat{i} + 2\hat{j} + \lambda\hat{k}$ and if $\left \frac{\vec{a} \cdot \vec{c}}{\vec{a} \cdot \vec{d}} - \frac{\vec{b} \cdot \vec{c}}{\vec{b} \cdot \vec{d}} \right = 0$, then $\lambda =$	6	-6	12	-1
141	Mathematics	The equation of the plane containing the line $\frac{x+1}{2} = \frac{y+2}{1} = \frac{z-2}{3}$ and the point $(1, -1, 3)$ is	$x - 2y - 3 = 0$	$2x + y - 1 = 0$	$3x - 2z + 3 = 0$	$2x$
142	Mathematics	The equation of the plane containing the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and perpendicular to the plane containing the lines $\frac{x}{2} = \frac{y}{3} = \frac{z}{1}$ and $\frac{x}{3} = \frac{y}{2} = \frac{z}{1}$ is	$x - 13y + z = 0$	$13x - 8y + 5z = 0$	$13x - 8y + z = 0$	$13x$
143	Mathematics	The distance of the point $P(3, 4, 4)$ from the point of intersection of the line joining the points $Q(3, -4, -5), R(2, -3, 1)$ and the plane $2x + y + z = 7$ is	7 units	9 units	11 units	6 units
144	Mathematics	If the lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then the value of k is	$\frac{3}{2}$	$-\frac{3}{2}$	$\frac{9}{2}$	$-\frac{2}{9}$
145	Mathematics	The equation of the line passing through the point of intersection of $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$ and also through the point $(2, 1, -2)$ is	$\vec{r} = (-\hat{i} - \hat{j} - \hat{k}) + \lambda(\hat{i} + 2\hat{j} + \hat{k})$	$\vec{r} = (-\hat{i} - \hat{j} + \hat{k}) + \lambda(2\hat{i} + 2\hat{j} + \hat{k})$	$\frac{x+1}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$	$\frac{x-3}{3} = \frac{y+1}{1}$
146	Mathematics	The graph with correct feasible region of L.P.P. for the constraints $2x + y \leq 10, y \leq x, y \geq 2, x, y \geq 0$ is ...	[Figure]	[Figure]	[Figure]	[Figure]
147	Mathematics	Bag I contains 3 red and 2 green balls and Bag II contains 5 red and 3 green balls. A ball is drawn from one of the bag at random and it is found to be green. Then the probability that it is drawn from Bag I is	$\frac{8}{31}$	$\frac{12}{31}$	$\frac{14}{31}$	$\frac{16}{31}$
148	Mathematics	In a game a man wins Rs. 40 if he gets 5 or 6 on a throw of a fair die and loses Rs. 20 for getting any other number on the die. If he decides to throw the die either till he gets a five or six or to a maximum of three throws, then his expected gain/loss (in rupees) is	-10	10	0	1
149	Mathematics	Let mean and standard deviation of probability distribution $\left[\begin{array}{c} X = x \\ P(X = x) \end{array} \right]$ be μ and σ respectively and if $\sigma - \mu = 2$ then $\sigma =$	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{9}{2}$
150	Mathematics	A fair coin is tossed 100 times. The chance of getting a head even number of times is	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{1}{2}$	$\frac{3}{4}$